1842, because of cost problems, and because of Babbage spending too much time redesigning the Analytical Engine.

1842—1843
If you've heard of Babbage, you've probably heard of Ada Lovelace, who is referred to by various names. (The Right Honourable Augusta Ada, Countess of Lovelace; Ada Lovelace; Augusta Ada Byron, Ada Byron, and some permutations...) She wrote to Babbage about a plan for how the Difference Engine might calculate Bernoulli numbers. This plan is now regarded "the first computer program."

There is a long and fascinating story about the relationship (no, it was not amorous) between Babbage and Lovelace; visit **www.hwwilson.com/print/leaders_info_TBernersdlee.htm** for a comprehensive, readable account.

Later...

More work on the Engines followed, and all of it was cancelled—only the original segment we referred to earlier was actually built. Babbage's ideas never made it to the factory.

In 1989—91, in true-blue English-heritage spirit, a team at London's Science Museum used modern components that had the basic clumsy qualities of those that Babbage and Clement used. Playing around with the design for the Difference Engine, they found the machine does indeed work. Babbage stands tall, respected, redeemed.

1.4 The Mathematical Foundation

1848
This is another of Britain's contributions to the development of computing: British Mathematician George Boole devised binary algebra, soon called Boolean algebra. This was essential for a